



Northeastern University

Optimizing Fund Allocation for Emission Reduction

Akash Alaparthi, Connor Breslin, William Geary, Ryan Jiang, Patrick Medek, and Nora Mehler

MATH 4025: Applied Mathematics Capstone

Professor John Lindhe

April 16, 2025

Abstract

Our project aims to find an optimized strategy for reducing carbon emissions within the energy and transportation sectors. Our analysis considered two vehicle types - internal combustion engine (ICE) and electric (EV); three vehicle classes - sedans, SUVs, and trucks; and three sources of power - natural gas, solar, and land-based wind. We calculated the lifetime emissions associated with each vehicle and method of energy production, as well as the subsidies required to change the current power grid composition and vehicle breakdown to alternative blends of renewables and EVs. Our optimization relied on matrices of reductions in CO₂ emissions resulting from changes in the composition of the energy grid (phasing out coal and natural gas in favor of renewables) and from the adoption of EVs, and a similar matrix to model the necessary subsidy costs. We found that with our flagship scenario, scenario 3 (modeled by a modified random walk in which the government decreases coal with a weighting of 0.75 and natural gas with a weighting of 0.25), at small and extremely large levels of government expenditure, most funding should be allocated towards EV adoption. Whereas, in the intermediate levels of government funding, nearly all government expenditure should be on renewable electricity generation.

	expenditure	ev_cost	egen_cost	total_reduction	ev_reduction	egen_reduction	pct_ev_cost	pct_egen_cost
0	\$5,000,000,000	\$3,443,708,545	\$1,010,757,612	-12,232,568,679	-7,704,374,578	-4,528,194,102	77.309%	22.691%
1	\$10,000,000,000	\$8,609,271,363	\$1,010,757,612	-23,789,130,546	-19,260,936,444	-4,528,194,102	89.493%	10.507%
2	\$25,000,000,000	\$22,384,105,545	\$1,010,757,612	-54,606,628,856	-50,078,434,754	-4,528,194,102	95.680%	4.320%
3	\$50,000,000,000	\$1,721,854,273	\$47,505,607,771	-127,093,442,600	-4,053,918,473	-123,039,524,127	3.498%	96.502%
4	\$100,000,000,000	\$5,165,562,818	\$94,404,760,974	-292,397,745,902	-12,959,653,419	-279,438,092,483	5.188%	94.812%
5	\$200,000,000,000	\$15,496,688,454	\$184,362,188,455	-590,545,632,655	-42,989,932,944	-547,555,699,711	7.754%	92.246%
6	\$350,000,000,000	\$99,867,547,815	\$250,061,433,245	-1,049,926,763,183	-297,241,849,116	-752,684,914,067	28.539%	71.461%
7	\$500,000,000,000	\$246,225,160,992	\$252,082,948,469	-1,496,350,153,382	-735,059,649,005	-761,290,504,376	49.412%	50.588%
8	\$750,000,000,000	\$497,615,884,802	\$252,082,948,469	-2,246,830,634,184	-1,485,540,129,808	-761,290,504,376	66.375%	33.625%

Figure 1: Fund distribution at different levels of government expenditure

Table Of Contents

I. Introduction.....	3
II. Data Collection.....	5
III. Methods.....	7
IV. Results.....	15
V. Limitations and Future Work.....	22
VI. Conclusion.....	25
References.....	28
Appendix A.....	32
Appendix B.....	33

I. Introduction

This project aims to look at emissions reduction holistically; we consider investment in electric vehicles and different sources of renewable energy, and account for both the total lifetime emissions associated with renewable energy generation and the source of electricity used to power EVs. Furthermore, by considering the subsidies necessary to switch from the current composition of renewable and nonrenewable energy and ICE and electric vehicles, the aim of this project is to find the appropriate composition of subsidies for EV adoption and renewable energy generation at a given level of government funding.

In 2022, the United States released 6,343 million metric tons of greenhouse gasses into the atmosphere; 79.7% of that was carbon dioxide, and the primary source of the emissions was the burning of fossil fuels (EPA, 2025). Transportation was responsible for 28% of direct greenhouse gas emissions in the United States, making that sector the most significant contributor to direct emissions (EPA, 2025). The importance of reducing carbon emissions to mitigate the effects of climate change has led to a push for electric vehicle adoption and investment in renewable energy, but determining the optimal strategy is challenging, given financial constraints and the emissions associated with the manufacture and operation of renewable energy technology.

Numerous studies, such as the Driving Cleaner report from the Union of Concerned Scientists, show that electric vehicles release significantly less carbon dioxide than traditional gas vehicles over their lifetimes (Reichmuth, et al., 2022). The negative correlation between EV sales and particulate matter emissions from vehicles is shown below in a graph constructed using data from Statista.

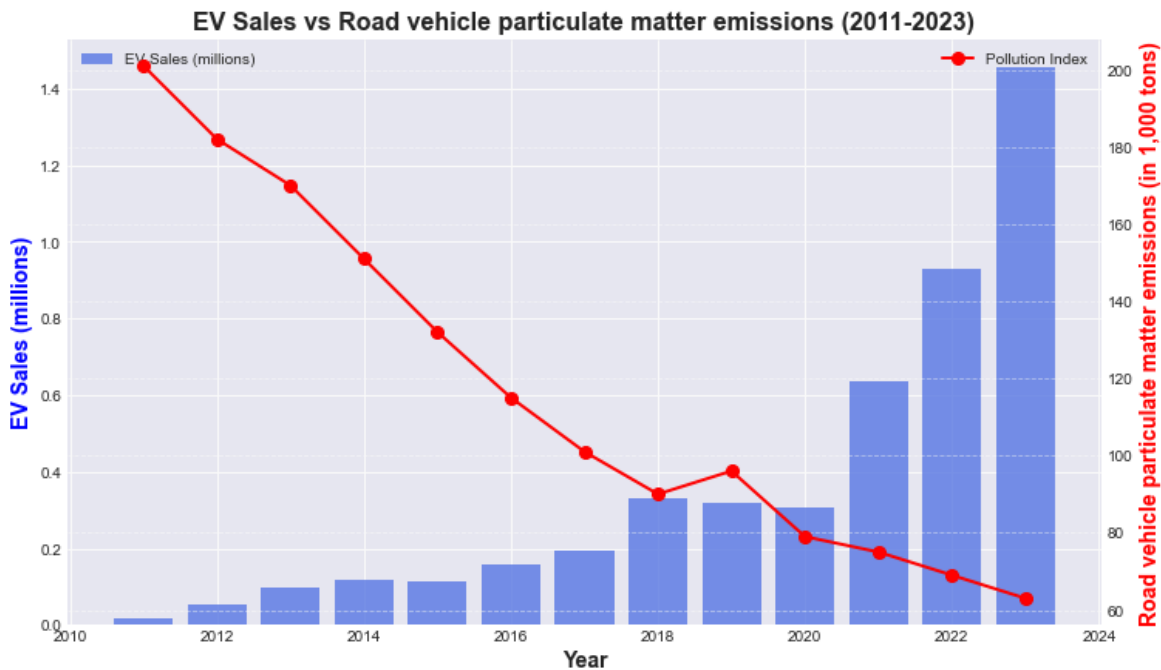


Figure 2: Correlation between road particulate matter and EV sales

Similarly, the negative correlation between renewable energy generation (Land-based Wind and Solar) and carbon dioxide emissions from power generation can be shown using data from the Congressional Budget Office. The graph constructed below depicts how switching to renewable energy sources, such as solar and wind, significantly impacts the CO₂ generated from producing electricity. As seen in the graph, coal's early dominance led to high emissions, but as renewables grew and coal's share shrank, emissions dropped noticeably. This highlights the strong role renewable energy sources play in reducing CO₂, even with natural gas and coal still in the mix.

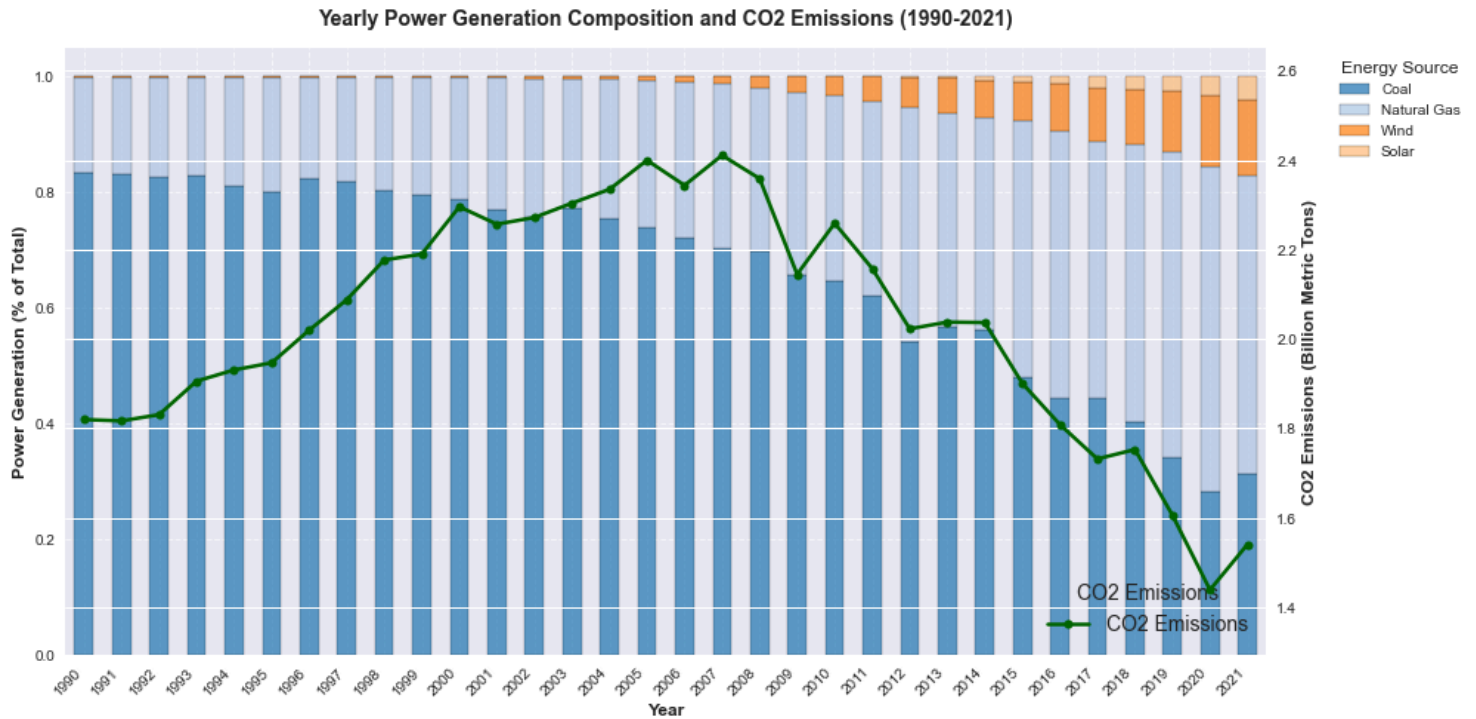


Figure 3: Correlation between composition of the power grid and CO₂ emissions

II. Data Collection

For this analysis, we gathered data from a wide range of sources, including government agencies, industry reports, academic research, and real-world datasets. Our goal was to ensure accuracy, comprehensiveness, and relevance when modeling vehicle emissions, energy consumption, and policy costs. Key sources included the Environmental Protection Agency (EPA), Energy Information Administration (EIA), and Department of Energy (DOE) for fuel economy estimates, energy production statistics, and emissions data, as well as the National Renewable Energy Laboratory (NREL) for information on the operation and maintenance costs for wind and solar plants. Additionally, we incorporated datasets on vehicle production materials and their associated emissions to capture the full lifecycle impact of different vehicle types. We

also wanted to find datasets with state-level data to examine regional trends in electric vehicle sales, energy production and cost, and sources of carbon emissions.

Due to the abundance of available data, we focused on sources that were the most reliable, relevant, and up-to-date. Some sources contained overlapping or conflicting data, which required us to prioritize datasets from government sources or peer-reviewed studies over industry reports with potential biases (for instance, the Tesla quarterly shareholder report). Additionally, some datasets lacked sufficient granularity for our analysis, leading us to exclude them in favor of more detailed sources. We also compiled the weights and gas mileage for the vehicles we chose to represent each of our nine categories, as outlined in the methods section, to consolidate data from different sources for use in our analyses and visualizations. Data regarding fuel consumption was primarily sourced from FuelEconomy.gov, which provides city, highway, and combined MPG (or MPGe for electric vehicles) for all new vehicles sold in the United States. Vehicle testing is conducted by the EPA's National Vehicle and Fuel Emissions Laboratory, and data for vehicle weights was sourced directly from each respective manufacturer's website.

Much of the intricacy of this project was due to the many different sources of data we would need in order to run our optimization, which would be from different sources, different methodologies, and different snapshots in time. In order to have consistent data, we strived to use raw inputs from as few different sources as possible, deferring to government agencies when possible, and for the intermediate metrics, to derive them ourselves.

III. Methods

Primary data on electricity generation and carbon emission weights was collected from the EIA. Vehicles were categorized into three classifications and three types, resulting in a total of nine distinct categories; Sedan, Truck, and SUV classifications, and Internal Combustion Engine (ICE), Hybrid (H, or Plug-In Hybrid Vehicle – PHEV), and Electric Vehicle (EV, or Battery Electric Vehicle – BEV) types. A standardized vehicle was selected to represent each of the nine distinct categories. The selected vehicles were chosen to be close substitutes within their types and among the most popular within their categories.

Equations were then developed to model the lifetime emissions of the nine vehicle categories. A constant in the lifetime emission functions of all types was the initial manufacture emissions. Due to the lack of reliable data directly from manufacturers, emissions were modeled using a standard vehicle chassis (Rodrigue, 2024), which was scaled by weight to match the size of each vehicle. This allowed for the estimation of the kilograms of each material used in the car chassis. Carbon emission values were then assigned to each unit of material in the chassis to calculate manufacturing emissions. This process was repeated for all vehicles we analyzed. For electric vehicles, an additional manufacturing emissions value for the battery was calculated using the same method.

To model the expected lifetime of each vehicle, the Weibull function—a commonly used approach for vehicle lifespans—was applied. Values for alpha and beta for each vehicle type were determined using an estimation of parameters random test, by finding lifespan data for EVs and ICE vehicles and then doing a regression to a standard Weibull curve. Then, using the expected value function of the Weibull equation, the expected lifetime of each vehicle type was

calculated. In order to model the expected lifetime emissions of ICE, H, and EV-type vehicles, the following functions were used:

$$ICE = \frac{C_1}{E(X_1)} + (MPG)^{-1}(MPY)(OilCoef) \quad (1)$$

$$H = \frac{C_2}{E(X_2)} + (MPG)^{-1}(MPY)(OilCoef) \quad (2)$$

$$EV = \frac{C}{E(X)} + (MPE)^{-1}(MPY)(\frac{dR}{dr}) \quad (3)$$

The following variables were used:

- C : Initial manufacturing emissions
- $E(X)$: Expected number of years until failure
- MPG : Miles per gallon of each vehicle
- MPY : Miles driven per year
- $OilCoef$: Amount of carbon emissions per gallon of oil consumed
- MPE : Miles per unit of electricity
- dR/dr : Carbon emissions of the electric grid per unit of energy.

Using the equations outlined above, we plotted the following diagrams: the Weibull Probability Density Function (PDF) and Cumulative Distribution Function (CDF) for ICE, H, and EV vehicles. The PDF curves illustrate that ICE vehicles have a higher probability of failure earlier in their lifespan compared to Hs and EVs, indicating a shorter expected lifetime. This

trend is further reinforced by the CDF, which shows that ICE vehicles accumulate failure probability more rapidly. In contrast, EVs exhibit a more gradual failure distribution, suggesting a longer lifespan. These lifetime estimates are crucial for determining total emissions, as vehicles with longer lifespans spread their manufacturing emissions over more years, ultimately influencing their impact on the environment.

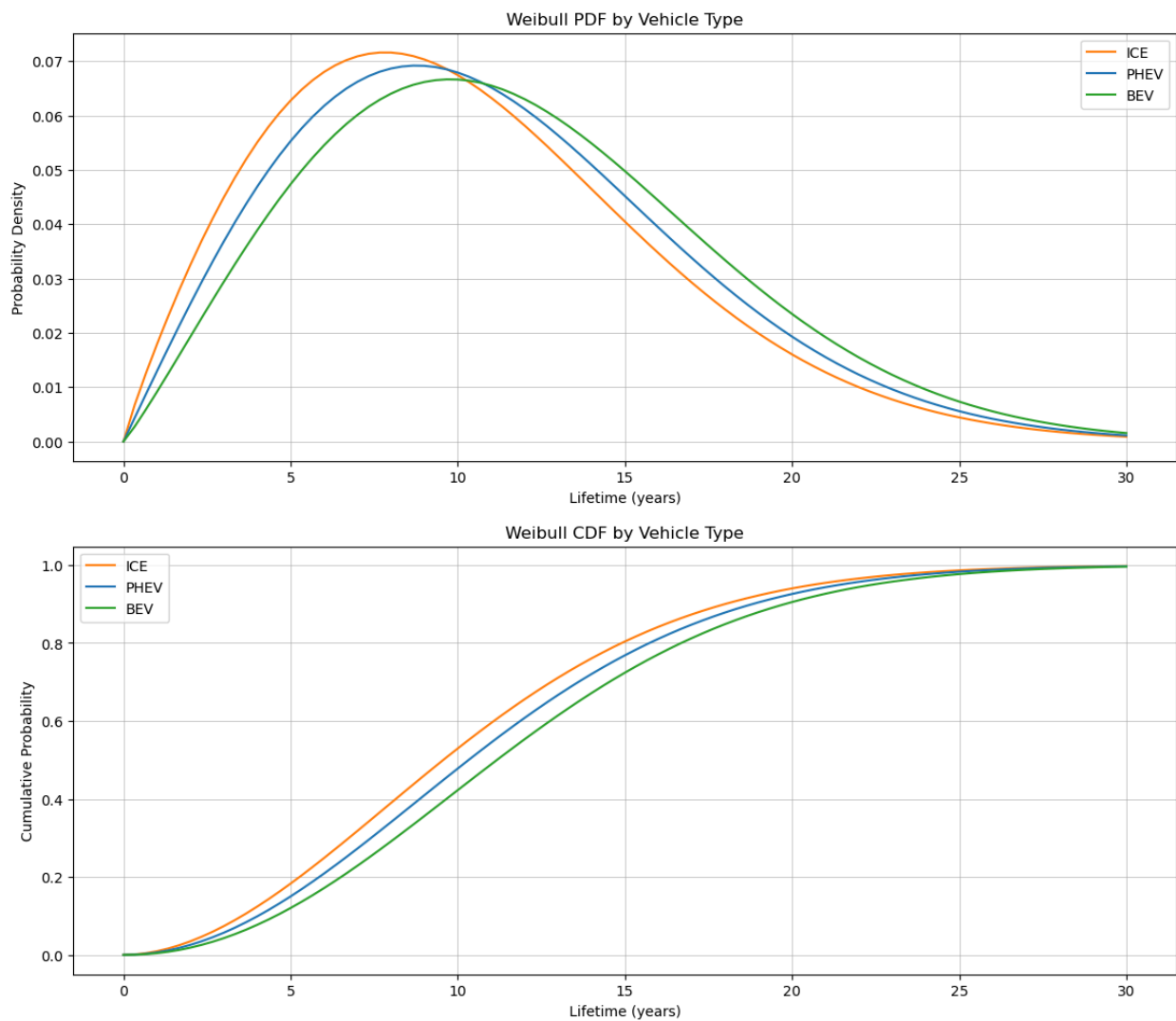


Figure 4: Weibull PDF and CDF for ICE vehicles, PHEVs, and EVs

The carbon emissions of the electric grid are dependent on the composition of the energy generation grid, which is influenced by the proportion of renewables integrated into the system. To model the electric grid, a generalization was made by representing a blend of land-based wind and solar energy as a single generic renewable source due to the similar costs of both technologies. A function to model the lifetime emissions of this renewable blend of energy generation was created, resembling the structure used for vehicles, with an initial coefficient for manufacturing emissions as shown below:

$$R = \frac{C}{E(X)} - (Scale)(Coef\ of\ R) \quad (4)$$

The following variables were used:

- C : Initial manufacturing emissions
- $E(X)$: Expected lifespan
- $Scale$: Current grid size in kWh
- $Coef\ of\ R$: Weighted average carbon intensity of electric grid

To compare each of the above equations to their per-unit cost in dollars, the existing ICE vehicle was taken as the baseline. The consumer was assumed to be rational and a profit maximizer, such that they would be indifferent between ICE, H, and EV-type vehicles after an appropriate subsidy. The subsidy for each of the three vehicle types is derived as a series of cash flows over the relevant time period, with an assumed 6.7% APR fully amortizing a 5-year car loan representing the car capitalization. The risk-free rate was taken at 4.3%, continuously

compounding (Based on 10-year T-Bond rate). The assumed cost per gallon is \$3.20, and the assumed cost per kWh is \$0.25 (AAA, 2015). These two prices were then assumed to grow at the risk-free rate in order to model an arbitrage-free forward curve. In addition, vehicle maintenance costs were considered based on a \$0.066 cost per mile of maintenance for an ICE vehicle and a \$0.042 cost per mile for an EV (AAA, 2015). Finally, ICE insurance costs per mile were calculated at \$0.10, and EV insurance costs were calculated at \$0.12 per mile (AAA, 2015). These costs were then set to grow at the risk-free rate. The cash flows were then calculated as an amortizing loan for 5 years for the difference between an ICE and EV vehicle as a cash outflow per month, the dollars spent per month in EV insurance vs ICE insurance costs, the dollars saved per month in EV charging costs vs ICE fueling costs, and the dollars saved per month in EV maintenance vs ICE maintenance. The values of all these nominal dollar terms were brought back to the initial time period at the appropriate consumer discount rate to calculate the present value of this stream of cash flows. This allowed for the calculation of the appropriate subsidy needed to ensure that the payoffs of all three of the vehicle types were zero, or if a subsidy was unnecessary, nonexistent. The function “ R ”, outlined above, for the dollar cost per kWh of generating the renewable blend, was then established to relate emissions reductions from increasing renewable expenditure and vehicle selection.

We began our comparison of energy sources—wind, solar, and natural gas—by calculating the levelized cost, or present value, of their generation. For a given power system, the levelized cost of energy is the ratio of the total lifetime investment, operational, and maintenance costs to the amount of energy produced (“Levelized Cost of Energy [LCOE]”, 2015). To calculate the levelized cost, we used the following equation:

$$LCOE = \frac{\sum_{t=1}^n \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}} \quad (5)$$

The following variables were used:

- n : Lifespan of the energy system (in years)
- I_t : Investment expenditure in year t
- M_t : Operation and maintenance expenditure in year t
- F_t : Fuel expenditure in year t (this was 0 for wind and solar)
- E_t : Electricity generation in year t
- r : Discount rate

The LCOE was then calculated for three different energy sources: Natural Gas, Land-based Wind, and Solar. Throughout the entire set of calculations, the risk-free rate was taken at 4.3% continuously compounding, the long term rate of inflation at 2.5% continuously compounding, and the number of hours in a year at 8766. For calculating the LCOE of natural gas, the correct Weighted Average Cost of Capital was found by using the market capitalization-weighted WACC of US-based Natural Gas companies at 5.39% continuously compounding, and the cost of debt at 4.95% continuously compounding (“Cost of Equity and Capital [US]”, 2025). The average lifespan of a natural gas plant was assumed to be 30 years, its initial cost at \$820, Maintenance and Operations cost at \$0.00428/kWh, and Fuel costs of \$2.19/Million BTUs (“Average Power Plant Operating Expenses...”; “U.S. construction costs...”, 2024). The financing cost was then found as a coupon bond with a par value of the initial cost, \$820. This

stream of cash outflows was then discounted back to the present with the WACC. The Maintenance cost was modeled by taking the per kWh maintenance cost and multiplying it by the number of hours in a year and having those costs grow at the estimated future rate of inflation. These figures were then discounted back into present value terms by discounting with the WACC. Fuel costs were then estimated by taking the Henry Hub spot price and normalizing its quantity to a per kWh rate. After, in order to approximate the forward curve, the spot price was to increase at the risk-free rate over the time period. This cost was then discounted back into present value terms by the WACC. Since all previous calculations were done in terms of kWh, modeling energy output was as simple as a sum of the discount factors over the plant's 30-year lifetime multiplied by the capacity factor for natural gas, 58.8% (“Capacity factors for selected energy sources...”, 2024). From here, the LCOE was found as a simple ratio of the present value of all future costs and the present value of future energy production as \$0.04142/kWh or \$41.42/MWh.

For calculating the LCOE of land-based wind, the correct Weighted Average Cost of Capital was found by using the market capitalization-weighted WACC of US-based Renewable Energy companies at 6.29% continuously compounding and the cost of debt at 6.41% continuously compounding (“Cost of Equity and Capital [US]”, 2025). The average lifespan of a land-based wind plant was assumed to be 20 years, its initial cost at \$1451, Maintenance and Operations cost at \$0.00239/kWh, and Fuel costs of \$0 (“Average Power Plant Operating Expenses...”; “U.S. construction costs...”, 2024). The financing cost was then found as a coupon bond with a par value of the initial cost, \$1451. This stream of cash outflows was then discounted back to the present with the WACC. The Maintenance cost was modeled by taking the per kWh maintenance cost and multiplying it by the number of hours in a year and having

those costs grow at the estimated future rate of inflation. These figures were then discounted back into present value terms by discounting with the WACC. Since all previous calculations were done in terms of kWh, again, modeling energy output was simply the sum of the discount factors over the plant's 20-year lifetime multiplied by the capacity factor for wind, 33.5% (“Capacity factors for selected energy sources...”, 2024). Similar to natural gas, the LCOE was found as a ratio of the present value of all future costs and the present value of future energy production as \$0.05331/kWh or \$53.31/MWh.

For calculating the LCOE of Solar, the correct Weighted Average Cost of Capital was found by using the market capitalization-weighted WACC of US-based Renewable Energy companies at 6.29% continuously compounding, and the cost of debt at 6.41% continuously compounding (“Cost of Equity and Capital [US]”, 2025). The average lifespan of a solar plant was assumed to be 33 years, its initial cost at \$1588, Maintenance and Operations cost at \$0.001939/kWh, and the Fuel costs of \$0 (“Average Power Plant Operating Expenses...”; “U.S. construction costs...”, 2024). The financing cost was then found as a coupon bond with a par value of the initial cost, \$1588. This stream of cash outflows was then discounted back to the present with the WACC. The Maintenance cost was modeled by taking the per kWh maintenance cost and multiplying it by the number of hours in a year and having those costs grow at the estimated future rate of inflation. These figures were then discounted back into present value terms by discounting with the WACC. The energy output was modeled by summing the discount factors over the plant's 33-year lifetime multiplied by the capacity factor for solar, 23.3% (“Capacity factors for selected energy sources...”, 2024). Finally, the LCOE was found as a ratio of the present value of all future costs and the present value of future energy production as \$0.06849/kWh or \$68.49/MWh.

IV. Results

We found that the breakeven subsidy required to incentivize consumers to switch from internal combustion engine (ICE) vehicles to electric vehicles (EVs) varies significantly by vehicle type. Based on cost assumptions for fuel, electricity, maintenance, and financing, the present value of cost savings needed to make EV adoption financially equivalent to ICE ownership is estimated at **\$11,045.17 for sedans, \$20,759.66 for SUVs, and \$15,174.49 for trucks**. These values represent the subsidy required to offset the lifetime cost difference between ICE and EV ownership. As detailed in the methods section, gas prices, electricity rates, maintenance, and financing costs all play a major role in determining the total cost of vehicle ownership. The analysis highlights that larger vehicles, such as SUVs and trucks, require significantly higher subsidies to reach cost parity with their ICE counterparts, suggesting that targeted incentives may be necessary to encourage widespread EV adoption across different vehicle classes.

We additionally found the LCOE for our different energy sources, computed from the process outlined in the methods and using publicly available LCOEs for sources we could not calculate. We then set out to begin our optimization using the equations originally outlined in the methods. We choose to do a two-layered numerical optimization. The first step of this process was to find the current allocation of energy generation for the US energy grid. We then set up around 500 different scenarios, representing every possible energy allocation for the energy grid, following a straight-line scenario where, as the share of renewables increases, the fossil fuel energy sources fall in proportion with their current market share. From all of these possible allocations, we calculated the weighted average LCOE for the energy grid, and then multiplied by the current size of the US energy grid to find the additional cost (or subsidy) required in order

to move from the current energy mix to every possible modeled energy mix. Since the LCOE is a yearly cost, and the comparison cost for the breakeven subsidy is a cost over the lifetime, we modeled the yearly cost as an annuity and used a cost of debt of 4.3% (this is the same as the risk-free rate as the government is assuming the risk of these projects, thus we can use the default-free cost of debt) continuously compounding in order to find the present value of the cost of obtaining every possible energy mix. We then found the carbon reduction from the current state to every possible state that we created, from here we divided out the marginal carbon reduction by the marginal cost in order to create a marginal cost curve to describe the number of dollars required to reduce emissions.

We considered three different scenarios, 1) where the government can perfectly phase out the dirtiest energy source (coal) first in favor of renewables, 2) where the government has no control over what fossil fuels are phased out first and all fossil fuels linearly decrease according to their pre-existing proportion, and 3) a modified random walk where the government is able to influence composition with a 0.75 chance of reducing coal and 0.25 chance of reducing natural gas. (This function is influenced by the bid-ask spread in Glosten and Milgrom 1985). The first scenario was created by following our initial process, starting with the initial array of the weights of electricity generation in the US and having coal be the first energy source swapped in for renewables. When all coal was phased out, next to go was natural gas, with every scenario being a 1/10th of 1% change in the overall energy mix. The second scenario was created by having all of the different fossil fuel energy sources falling in accordance with their weighting in the initial distribution such that every iteration again had a 1/10th of 1% change in the energy mix. The third scenario was created as a middle ground, where the government had some influence on the fossil fuel that was being phased out for renewables. It was modeled off a modified random

walk-in with $p = 0.75$ of the coal decreasing and $1-p = 0.25$ of natural gas decreasing by 1/10th of 1%.

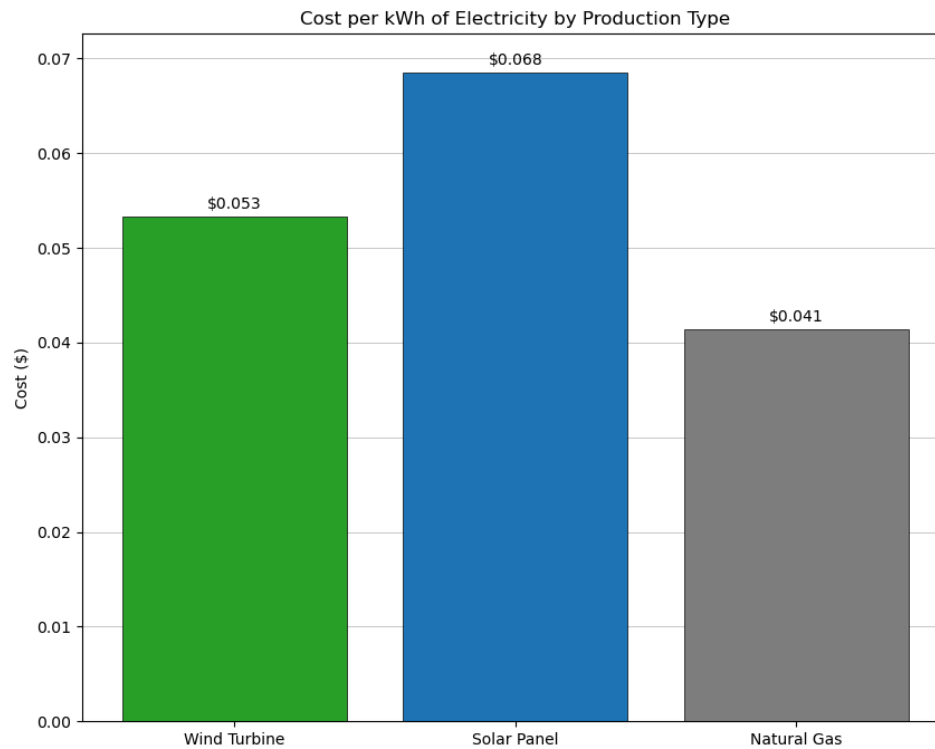


Figure 5: Electricity cost per kWh by source

The crux of the project was the acknowledgment from Equation 3, that the emissions reduction derived from an electric vehicle is a function of the electricity composition of the electric grid. This is because if the energy produced from the electric grid is cleaner, electric vehicles are inherently cleaner, too. We then calculated the lifetime emissions of an ICE vehicle as defined in equation 1, which is a constant 55,160.7 kg of carbon/vehicle for sedans and SUVs—given their equivalent MPG values—and 84,192.63 kg of carbon/vehicle for trucks. Hybrid vehicles were not included in this calculation given the model’s inability to decipher why they were not purchased, despite price ranges being similar to those of ICE counterparts. This

phenomenon is explained further in Limitations and Future Work. The production-level carbon emissions were factored into the delta carbon emissions between ICE and EV vehicles later. We then took every electricity composition scenario that we created and found the lifetime carbon emissions for an EV from Equation 3. Following this calculation, we took the difference in ICE and EV lifetime emissions, additionally subtracting out the production emissions in order to find the carbon savings per vehicle in each energy case. We then used the break-even subsidy values calculated above, which are constant values for each vehicle category (Sedan, SUV, Truck), to find the carbon reduction per dollar under every possible scenario in order to create a similar marginal cost curve.

Using the various energy generation scenarios, and the corresponding savings per EV vehicles, which are dependent on the energy grid composition, we could start to build the optimization model. Recalling the three energy reduction scenarios (coal-first reduction, equal reduction of energy generation types, random reduction of energy generation types), and the three vehicle categories (Sedans, SUVs, Trucks), as well as the overall weighted combination of all vehicle categories, there were nine combinations to be modeled and an additional three to be calculated. For each energy reduction scenario–vehicle category combination (ex. coal-first reduction and SUVs), four columns were copied from the model into a [CSV file](#): (c1) the cost of electricity generation at each iteration, (c2) the carbon reduction in kg CO₂ per dollar spent on electricity generation at each iteration, (c3) the count of electric vehicles at each vehicle iteration, and (c4) the carbon reduction in kg CO₂ in electric vehicles per dollar spent on electricity generation at each iteration. In each combination, two additional columns had to be calculated: (c5) the cost of subsidizing EVs at each quantity, which was determined by multiplying the count of electric vehicles column (c3) by vehicle category subsidy (Sedan = \$11,045.17, SUV =

\$20,759.66, Truck = \$15,174.49), or in the final case, the weighted average (\$17,218.54); and (c6) the total carbon reduction in kg CO₂ at each iteration of electricity generation, which was calculated by multiplying the reduction per dollar at each iteration (c2) by the cost at each iteration (c1). All calculations were completed in a [Jupyter Notebook file](#).

With these six columns, the optimization could be set up. To determine the optimal allocation of funds between clean energy generation and electric vehicle subsidies, a matrix of all reduction scenarios was created, with rows representing each iteration in changes to the energy grid composition, and columns representing the number of electric vehicles subsidized. To create this matrix, two sub-matrices were created, one representing total reductions in carbon emissions from electricity generation, and the other representing total reductions in carbon emissions from electric vehicles. To create the matrix of electricity generation reductions, the column (c6) of total reductions at each iteration was repeated over the number of scenarios of electric vehicles, as the composition of the energy grid is independent of the number of electric vehicles on the road. Conversely, to create the matrix of total EV reductions, the column (c4) of reductions from EVs (due to changes in the energy grid) was multiplied by the transpose of the column (c6) of the total cost of subsidizing the count of EVs at each iteration, thus representing the total carbon reductions among EVs, dependent on both the energy grid composition (rows) and the number of EVs that replaced their ICE counterparts (columns). In order to determine the total carbon reductions in each scenario, the two sub-matrices had to be added together, providing a matrix (m1) of total carbon reductions in kg CO₂ for each expenditure split between electricity generation and electric vehicle subsidies.

In order to determine the optimal solution that fell within a given level of government expenditure, a matrix of total costs also had to be constructed to filter out combinations that were

too expensive. This was done by creating two sub-matrices, one for electricity generation costs, and the other for EV subsidy costs, and summing the two together, similar to how the total emissions reductions matrix was constructed. To create the electricity generation cost sub-matrix, the column of cost for electricity generation at each iteration (c1) was repeated over the number of scenarios of electric vehicles. Similarly, to create the EV subsidy cost sub-matrix, the column of the cost of subsidizing EVs at each quantity (c5) was transposed, then repeated (as rows) over the number of iterations of electricity generation scenarios. The two matrices were summed together to get a total cost matrix (m2) for each expenditure split. The total reduction (m1) and total cost (m2) matrices represented the same scenarios among the rows and columns, i.e., the position (5, 24) represented the same scenario of expenditure split in m1 and m2, the only difference being that m1 represented total carbon reductions while m2 represented total costs.

Within the total cost matrix (m2), any value that was greater than the given expenditure level was set to \$0. Similarly, within the total carbon reduction matrix (m1), all combinations that were too expensive in the total cost matrix (m2) were set to 0 in the total reduction matrix (m1) as well, filtering out all carbon emissions reductions that were outside possible bounds set by the expenditure level. Once filtered, the index (combination) of the greatest carbon reduction was determined from the carbon reduction matrix (m1). Subsequently, this index was used in the cost sub-matrices to determine the optimal split of electricity generation and EV subsidy spending, and in the carbon emissions reductions sub-matrices to discover the share of carbon reduction between energy generation and EVs. The index was also used in the total cost matrix (m2) to determine the final level of government expenditure used to achieve the level of carbon emissions reduction. This step was necessary as we did not use continuous functions, but rather

iterative steps, so it was possible that the complete level of government expenditure was not used (although the values were often very close).

This process was repeated for each combination of energy reduction scenarios and vehicle categories, and was also completed for the weighted average of vehicle categories. Rather than assuming that all consumers would purchase a certain vehicle category (Sedan, SUV, or Truck) to substitute their ICE vehicle with an EV, we used the ratios of the most common selling vehicles on the road, for both ICE (Capparella, 2025) and EVs (Mathilde, 2025), to determine the ratio of sedans, SUVs, and trucks that consumers desire. Using the ratio of ICE (“How Many Cars ...”, 2025) and EVs (“Hybrid-Electric, Plug-in ...”, 2025) compared to the total number of vehicles on the road, we determined a rough split of sedans (15.8%), SUVs (48.3%), and trucks (35.9%) on the road today. These percentages were multiplied by the carbon reduction per dollar costs from sedans, SUVs, and trucks in each of the three energy reduction scenarios, providing a weighted average. Unlike before, the subsidy cost was now also the weighted average of the various sedan, SUV, and truck subsidies. Using this weighted average, the optimal expenditure split in each energy reduction scenario was calculated.

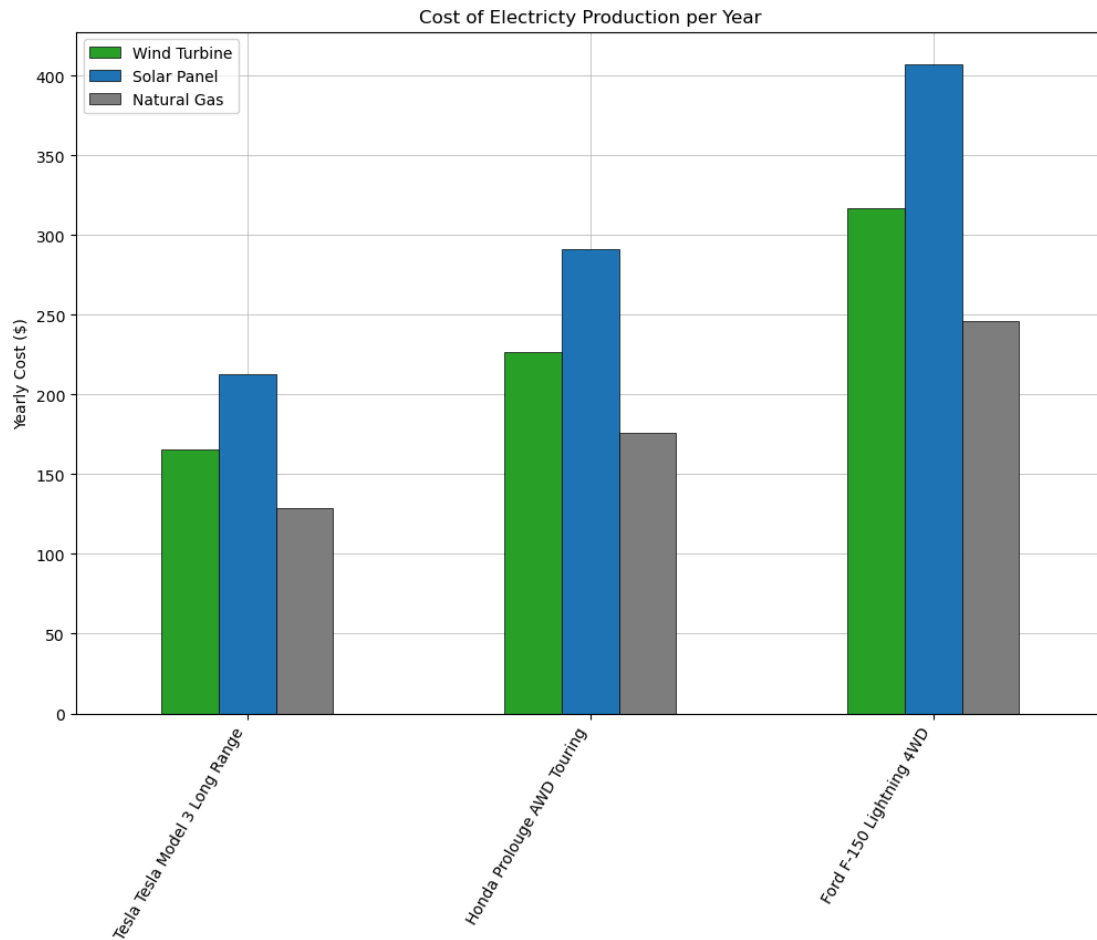


Figure 6: Cost of electricity per year for EV classes, by energy source

V. Limitations and Future Work

One of the main limitations of our project was the need to rely on assumptions and generalizations due to data constraints. Estimating lifetime vehicle emissions required assuming standardized vehicle lifespans, material compositions, usage patterns, and manufacturing emissions. In reality, these factors vary significantly across different models, driving conditions, and individual behaviors, introducing uncertainty into our calculations.

Similarly, determining the necessary subsidies for EV adoption involved estimating costs such as electricity rates, fuel prices, and financing terms, all of which fluctuate over time and

across regions, beyond our estimated inflation calculations. Additionally, insurance and maintenance costs were approximated on a per-mile basis despite substantial variation between individual drivers and vehicle types. These necessary simplifications mean that our subsidy estimates provide broad directional insights rather than precise policy recommendations. We also opted not to include hybrid vehicles in our final analysis, as hybrid vehicles would not fit within the 4-factor model we developed for calculating EV subsidies. They would require at least a 5th component of repair cost/time, which due to the timing of cash flows and replication concerns, is incredibly difficult to model in a replicating asset model.

In evaluating alternative emission-reduction strategies, we calculated the levelized cost of energy (LCOE) for different power sources. This required making assumptions about financing conditions, capacity factors, and long-term cost trends for fuel and maintenance. Because these variables change due to economic shifts, policy changes, and technological advancements, our LCOE estimates should be interpreted as general benchmarks rather than exact projections. Given the downward trend in renewable energy costs, these benchmarks are likely to remain comparatively elevated over time. Furthermore, our analysis did not account for location-specific factors such as grid infrastructure, regional energy policies, or market dynamics that could significantly impact the viability of different energy sources.

Overall, while our analysis provides a useful framework for comparing emission-reduction strategies, the necessary assumptions introduce inherent uncertainties. A more precise approach would require access to detailed, real-world data that goes beyond the scope of our project.

In the future, we would aim to refine our analysis by incorporating more detailed and granular data to reduce reliance on broad assumptions. A key improvement would be gaining

access to more specific consumer data, such as real-world driving patterns, vehicle depreciation rates, and individual fuel or electricity consumption. This would allow us to better model variations in lifetime emissions and costs across different types of drivers, vehicles, and regions. If possible, we would work with industry sources, government agencies, or insurance companies to obtain anonymized data on vehicle usage, maintenance costs, and financing behavior to improve subsidy estimations.

Additionally, rather than applying uniform assumptions across the entire US, we would conduct a more regionally focused analysis to determine the optimal budget allocation strategy and how budget allocation differs between the Northeast, Southeast, Midwest, etc. This would involve assessing variations in energy prices, infrastructure readiness for EV adoption, land availability, and regional emissions intensities of electricity generation. By segmenting consumers based on region, driving habits, and financial incentives, we could develop a more precise model for determining where subsidies or renewable energy investments would have the highest impact per dollar spent.

For the levelized cost of energy (LCOE) calculations, we would seek more granular market data on financing structures, evolving fuel prices, and technological improvements over time. By incorporating location-specific capacity factors and regional grid demand trends, we could refine our assessment of whether investment in renewable energy or EV subsidies would be more effective in a given area. Ultimately, with access to more advanced datasets and predictive modeling techniques, we could develop a more targeted and adaptive approach to minimizing emissions within a finite budget.

VI. Conclusion

The optimal solution to the level of government expenditure is highly dependent on the nature of the underlying assumptions. We considered three different scenarios, one where the government is able to perfectly phase out the dirtiest energy source (coal) first in favor of renewables, another where the government has no control over what fossil fuels are phased out first and all fossil fuels linearly decrease according to their pre-existing proportion, and thirdly, a modified random walk where the government is able to influence it with a 0.75 chance of reducing coal and 0.25 chance of reducing natural gas. We then used our optimization model described above in Results to create the optimal spending composition of EV adoption and renewable energy generation at each level of government spending as well as the carbon emission reduction associated with those values. We also provided a tabulated version of this cost blend data at key potential funding levels. We found that with our flagship scenario, scenario 3, at small and extremely large levels of government expenditure, most funding should be allocated towards EV adoption. Whereas, in the intermediate levels of government funding, nearly all government expenditure should be spent on renewable electricity generation.

	expenditure	ev_cost	egen_cost	total_reduction	ev_reduction	egen_reduction	pct_ev_cost	pct_egen_cost
0	\$5,000,000,000	\$3,443,708,545	\$1,010,757,612	-12,232,568,679	-7,704,374,578	-4,528,194,102	77.309%	22.691%
1	\$10,000,000,000	\$8,609,271,363	\$1,010,757,612	-23,789,130,546	-19,260,936,444	-4,528,194,102	89.493%	10.507%
2	\$25,000,000,000	\$22,384,105,545	\$1,010,757,612	-54,606,628,856	-50,078,434,754	-4,528,194,102	95.680%	4.320%
3	\$50,000,000,000	\$1,721,854,273	\$47,505,607,771	-127,093,442,600	-4,053,918,473	-123,039,524,127	3.498%	96.502%
4	\$100,000,000,000	\$5,165,562,818	\$94,404,760,974	-292,397,745,902	-12,959,653,419	-279,438,092,483	5.188%	94.812%
5	\$200,000,000,000	\$15,496,688,454	\$184,362,188,455	-590,545,632,655	-42,989,932,944	-547,555,699,711	7.754%	92.246%
6	\$350,000,000,000	\$99,867,547,815	\$250,061,433,245	-1,049,926,763,183	-297,241,849,116	-752,684,914,067	28.539%	71.461%
7	\$500,000,000,000	\$246,225,160,992	\$252,082,948,469	-1,496,350,153,382	-735,059,649,005	-761,290,504,376	49.412%	50.588%
8	\$750,000,000,000	\$497,615,884,802	\$252,082,948,469	-2,246,830,634,184	-1,485,540,129,808	-761,290,504,376	66.375%	33.625%

Figure 7: Optimized fund distribution at different levels of expenditure, scenario 3

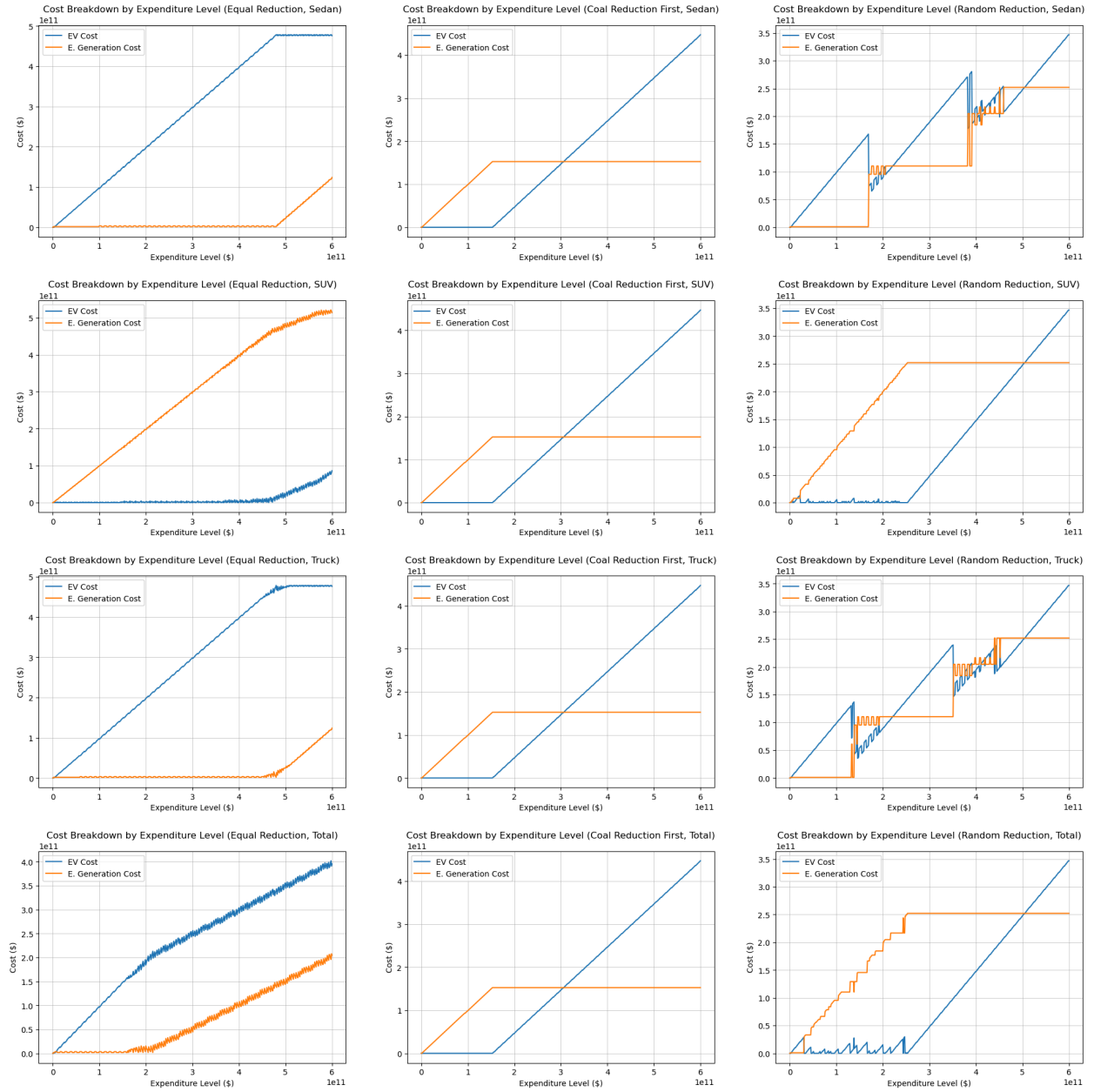


Figure 8: Cost breakdown for each fossil fuel reduction scenario and vehicle class, by expenditure

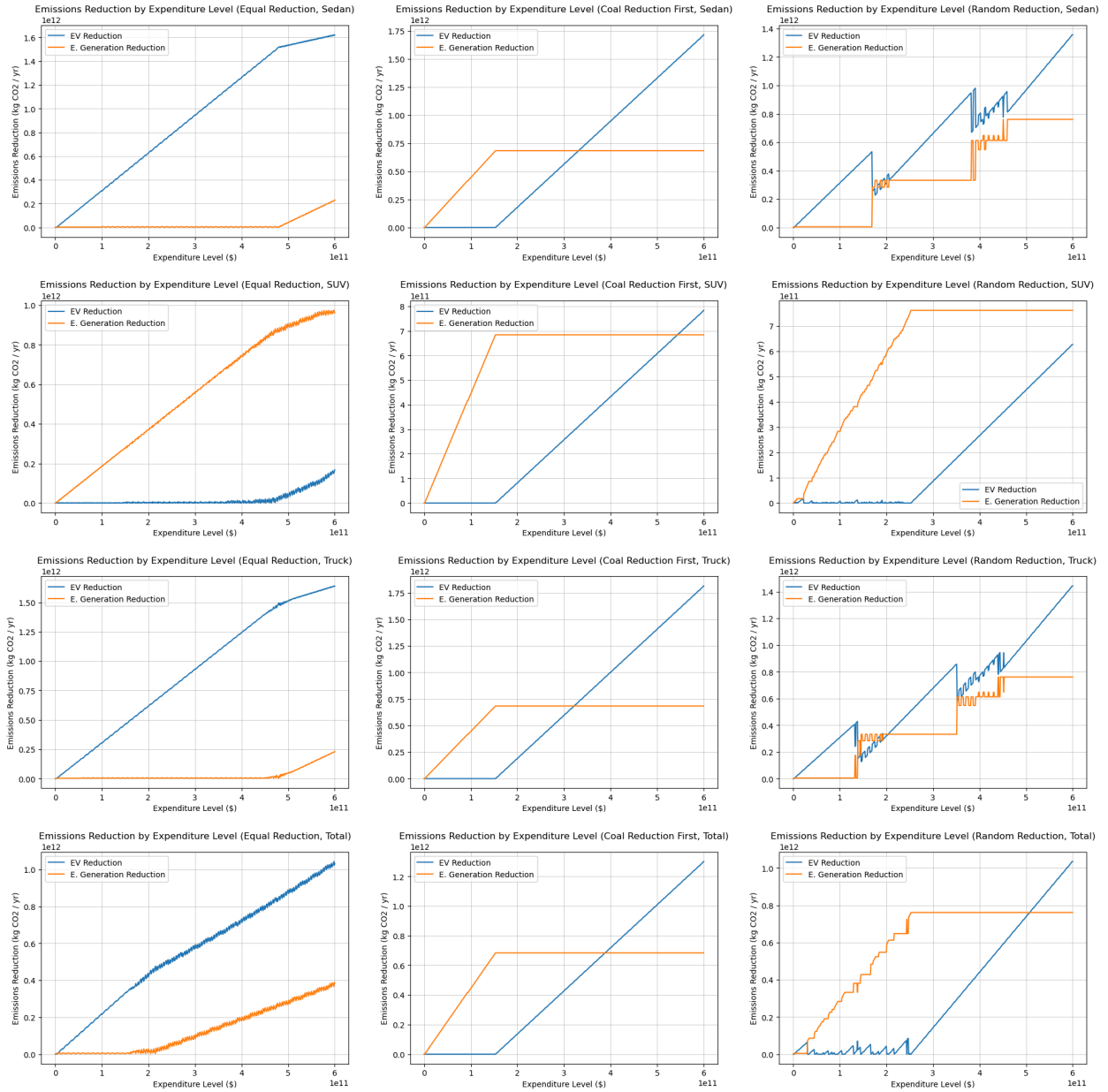


Figure 9: Emissions reduction breakdown for each fossil fuel reduction scenario, by expenditure

References

- “Average Power Plant Operating Expenses for Major U.S. Investor-Owned Electric Utilities, 2013 through 2023 (Mills per Kilowatthour).” U.S. Energy Information Administration, https://www.eia.gov/electricity/annual/html/epa_08_04.html.
- “Capacity factors for selected energy sources in the United States in 2023,” *Statista*, Apr. 2024, <https://www.statista.com/statistics/183680/us-average-capacity-factors-by-selected-energy-source-since-1998/>.
- Capparella, Joey. “The 25 Bestselling Cars, Trucks, and SUVs of 2023.” *Car and Driver*, Hearst Digital Media, 5 Jan. 2025, www.caranddriver.com/.
- Carrier, Mathilde. “Best-Selling Battery-Electric Cars in the United States 2024.” *Statista*, 18 Feb. 2025, www.statista.com/statistics/257966/best-selling-electric-cars-in-the-united-states/.
- “Cost of Equity and Capital (US).” *NYU*, January 2025. https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/wacc.html.
- “Electric Vehicles Registered in 2023.” *Alternative Fuels Data Center*, U.S. Department of Energy, https://afdc.energy.gov/transatlas#/?fuel=ELEC&view=vehicle_count.

“Emissions of Carbon Dioxide in the Electric Power Sector.” *Congressional Budget Office*, 13 Dec. 2022, <https://www.cbo.gov/publication/58419>.

“Global EV Outlook 2024”, *IEA*, April 2024,
<https://www.iea.org/reports/global-ev-outlook-2024>.

Hammond, Rob, and Aubryn Cooperman. “Windfarm Operations and Maintenance Cost-Benefit Analysis Tool (WOMBAT).” National Renewable Energy Laboratory, Oct. 2022.
<https://www.nrel.gov/docs/fy23osti/83712.pdf>.

“How Many Cars Are in the U.S.?” *Hedges & Company*, 17 Feb. 2025,
hedgescompany.com/automotive-market-research-statistics/auto-mailing-lists-and-marketing/.

“Hybrid-Electric, Plug-in Hybrid-Electric and Electric Vehicle Sales.” *Bureau of Transportation Statistics*, www.bts.gov/content/gasoline-hybrid-and-electric-vehicle-sales. Accessed 10 Apr. 2025.

“Inventory of U.S. Greenhouse Gas Emissions and Sinks.” *EPA*, Environmental Protection Agency, 15 Jan. 2025, www.epa.gov/ghgemissions/global-greenhouse-gas-overview.

Lawrence R. Glosten, Paul R. Milgrom, “Bid, ask and transaction prices in a specialist market with heterogeneously informed traders,” *Journal of Financial Economics*, Volume 14,

Issue 1, 1985, Pages 71-100, ISSN 0304-405X,
[https://doi.org/10.1016/0304-405X\(85\)90044-3](https://doi.org/10.1016/0304-405X(85)90044-3).

“Levelized Cost of Energy (LCOE).” *U.S. Department of Energy*, Aug. 2015,
<https://www.energy.gov/sites/prod/files/2015/08/f25/LCOE.pdf>.

Mcguire, Gavin. “Slow Charge Point Rollout Risks Stalling US EV Sales Momentum.” *Reuters*,
9 Oct. 2024,
www.reuters.com/markets/commodities/slow-charge-point-rollout-risks-stalling-us-ev-sales-momentum-maguire-2024-10-09/.

“Projected Costs of Generating Electricity 2020.” *International Energy Agency*, Dec. 2020,
<https://www.iea.org/reports/projected-costs-of-generating-electricity-2020>.

Reichmuth, D., Dunn, J., and Anair, D. “Driving Cleaner.” *Union of Concerned Scientists*, July
2022, www.ucsusa.org/sites/default/files/2022-07/driving-cleaner-report_0.pdf.

Rodrigue, Jean-Paul. “Material Components of a Car.” *The Geography of Transport Systems*, 6th
ed., 2024,
<https://transportgeography.org/contents/chapter4/transportation-sustainability-decarbonization/material-components-car/>.

“Transportation Sector Emissions.” *EPA*, Environmental Protection Agency, 16 Jan. 2025,
<https://www.epa.gov/ghgemissions/transportation-sector-emissions>.

“True Cost of Electric Vehicles.” *AAA Automotive*, October 7, 2015.
<https://www.aaa.com/autorepair/articles/true-cost-of-ev>.

Tucker, Sean. “America Set EV Sales Record in 2024.” *Kelley Blue Book*, 14 Jan. 2025,
www.kbb.com/car-news/america-set-ev-sales-record-in-2024/.

“U.S. Construction Costs Rose Slightly for Solar and Wind, Dropped for Natural Gas in 2022.”
Today in Energy, U.S. Energy Information Administration, 22 Oct. 2024,
<https://www.eia.gov/todayinenergy/detail.php?id=63485>.

“U.S. Plug-in Electric Light Vehicle Sales 2020.” *Statista*, 24 Oct. 2024,
www.statista.com/statistics/665823/sales-of-plug-in-light-vehicles-in-the-us/.

“U.S. Road Vehicle Particle Emissions 2022.” *Statista*, 15 Mar. 2024,
www.statista.com/statistics/1234827/road-transportation-of-particulate-matter-emissions-us/.

Appendix A

Additional Equations

$$f(x) = \frac{\beta}{\eta} \left(\frac{x}{\eta} \right)^{\beta-1} e^{-\left(\frac{x}{\eta} \right)^{\beta}}$$

$$F(x) = 1 - e^{-\left(\frac{x}{\eta} \right)^{\beta}}$$

PDF and CDF equations for the Weibull function that was used to determine the expected lifespan of vehicles, x , based on the parameters beta and eta, which differ between vehicle types.

Appendix B

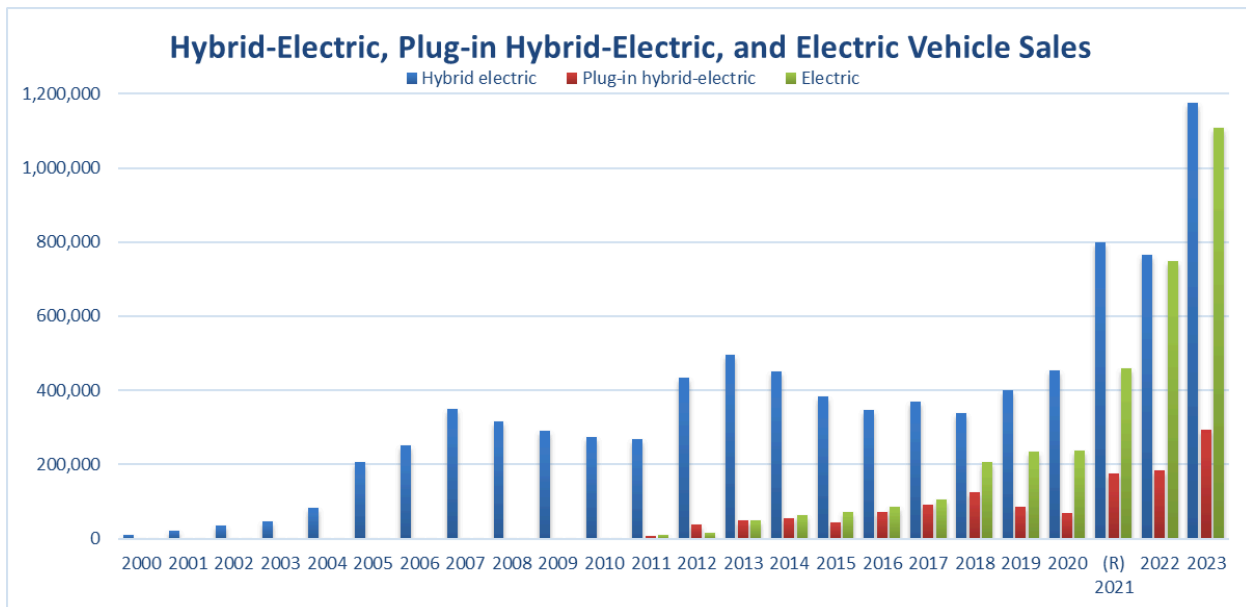


Figure 10: HEV, PHEV, and EV Sales from 2000-2023 (Bureau of Transportation Statistics)

“While Hybrid Electric Vehicles (HEVs) were first on the market, electric vehicles (EVs) have quickly exploded in popularity. Starting from 2021 in particular is interesting for showing a near doubling in sales for HEVs, EVs, and PHEVs.” (Bureau of Transportation Statistics)

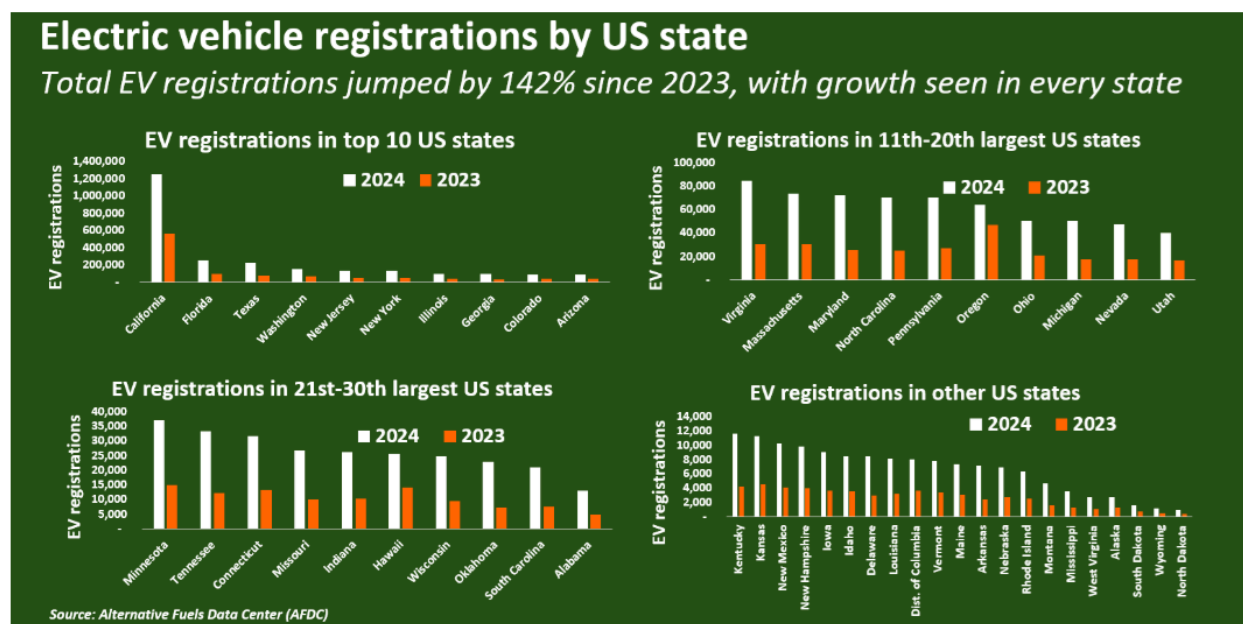


Figure 11: EV registrations by US state (as of September 2024)

U.S. EV registrations have grown by over 140% since early 2023, reaching 3.5 million in September 2024, with California, Florida, and Texas leading the market. Almost all states showed at least 100% growth since 2023, highlighting strong nationwide adoption. However, public charging infrastructure has only increased by 22%, creating a gap that could slow adoption despite rising sales. The future of EV adoption depends on government incentives and the success of the National Electric Vehicle Infrastructure (NEVI) program to expand charging access.

Automaker	Q1 2022	Q2 2022	Q3 2022	Q4 2022	Q1 2023	Q2 2023	Q3 2023	Q4 2023	Q1 2024	Q2 2024	Q3 2024	Q4 2024
Tesla	74.8	66.1	60.3	58.0	62.4	59.3	50.0	50.9	52.1	49.7	48.2	44.4
Ford	4.4	7.8	9.7	9.0	4.2	5.0	6.7	8.2	7.5	7.2	8.6	8.7
General Motors	0.3	3.7	8.0	7.2	8.0	5.3	6.1	6.1	6.0	6.6	9.3	12.6
Honda Motor Co	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	4.4	5.4
Nissan	2.5	1.7	0.7	1.5	2.0	1.4	1.9	1.6	2.0	2.2	2.9	2.5
Volkswagen Group	4.6	4.6	6.3	6.7	6.1	4.8	6.5	6.4	5.1	3.6	3.4	2.8
Hyundai Motor Group (incl. Kia)	8.9	9.1	5.2	4.3	5.6	7.3	9.9	8.0	8.5	11.2	6.8	11.4
Mercedes-Benz	1.2	1.0	1.4	2.5	2.0	3.1	3.3	3.4	4.6	2.8	2.7	1.2

Figure 12: U.S EV Market Share by Automaker (Q1 2022 - Q4 2024)

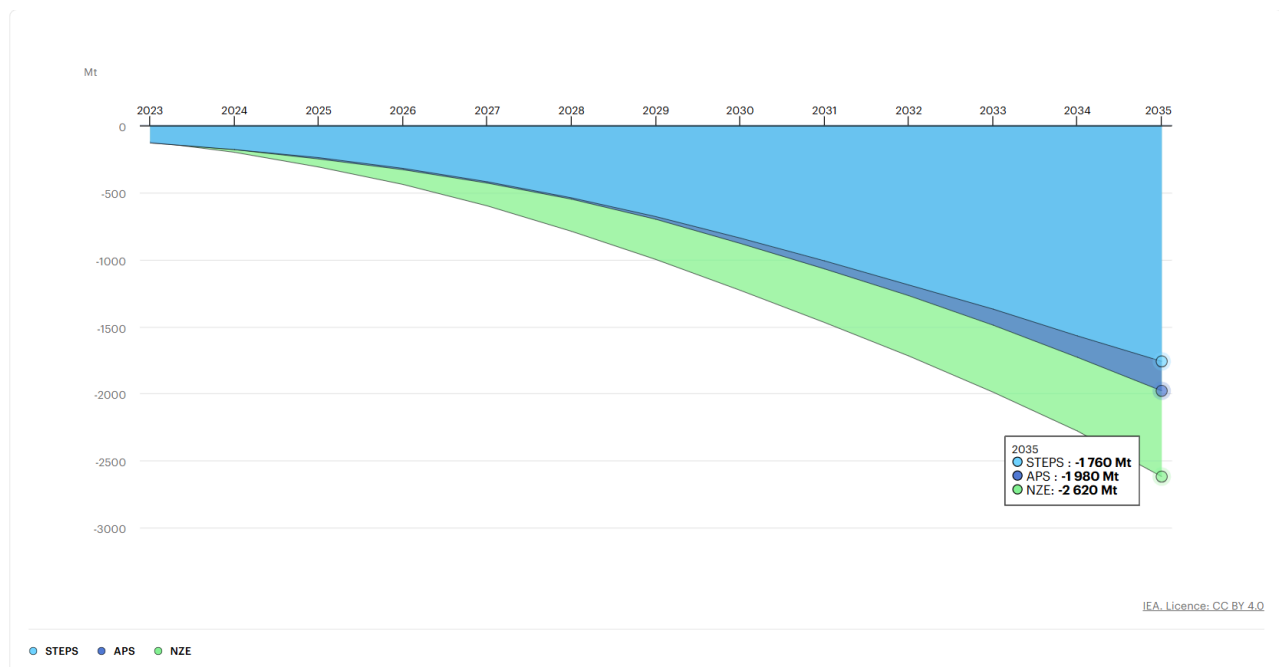


Figure 13: Net avoided well-to-wheel greenhouse gas emissions from electric vehicle deployment, 2023-2035

The chart shows how much greenhouse gas (GHG) emissions can be avoided worldwide from 2023 to 2035 if more electric vehicles (EVs) are adopted. It compares 3 different scenarios:

- (1) Stated Policies Scenario (STEPS): This scenario projects moderate growth in EV adoption based on current policies, with EVs making up a significant portion of new car sales by 2035. (around 25%)
- (2) Announced Pledges Scenario (APS): This scenario assumes countries meet their announced targets, leading to a higher rate of EV adoption compared to STEPS.
- (3) Net Zero Emissions by 2050 Scenario (NZE): This scenario requires rapid and widespread EV adoption to achieve net-zero emissions by 2050, with nearly all new car sales being electric by 2035.

The largest emission reductions are expected when both EV adoption increases and the electricity grid becomes cleaner. This shows that switching to EVs alone isn't enough—cleaner energy sources are also necessary for maximum environmental benefits.